14) Using RAW Sockets to implement packet Sniffing

Aim:

To implement a code using RAW Sockets to implement packet Sniffing.

Procedure:

Coding:

```
from Scapy.all import Sniff
from Scapy.layers.inet import IP, TCP, UDP, ICMP
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        def_ip = IP_layer.dst
    protocol_name = ""
    if protocol == 1:
        protocol_name = "ICMP"
    elif protocol == 6
        protocol_name = "TCP"
```

```
elif protocol == 17:
    protocol_name = "UDP"
else:
    protocol_name = "unknown protocol"
print(f"protocol: {protocol_name}")
print(f"Source Ip: {src_ip}")
print(f"Destination Ip: {dst_ip}")
print("-" * 50)
sniff(iface="wi-Fi", prn=packet_callback,
      filter="ip", store=0)
```

Output:

```
C:\Users\tkalal\Downloads>python packet.py
protocol: TCP
Source Ip: 57.155.141.117
Destination Ip: 192.168.1.6
- - -
protocol: TCP
Source Ip: 192.168.1.6
Destination Ip: 57.155.141.117
```

protocol : TCP

Source Ip : 192.168.1.6

Destination Ip : 57.155.141.117

protocol : TCP

Source Ip : 192.168.1.6

Destination Ip : 57.155.141.117

protocol : TCP

Source Ip : 57.155.141.117

Destinat Ip : 192.168.1.6

protocol : TCP

Source Ip : 57.155.141.117

Destinatio Ip : 192.168.1.6

protocol : UDP

Source Ip : 192.168.1.6

Destinatio Ip : 192.168.1.1

protocol : UDP

Source Ip : 192.168.1.6

Destinatio Ip : 192.168.1.1

protocol: UDP

Source Ip : 192.168.1.6

Destination Ip : 192.168.1.1

protocol : UDP

Source IP : 192.168.1.1

Destination : 192.168.1.6

Result:

Therefore, the RAW Sockets to implement packet Sniffer executed Successfully.